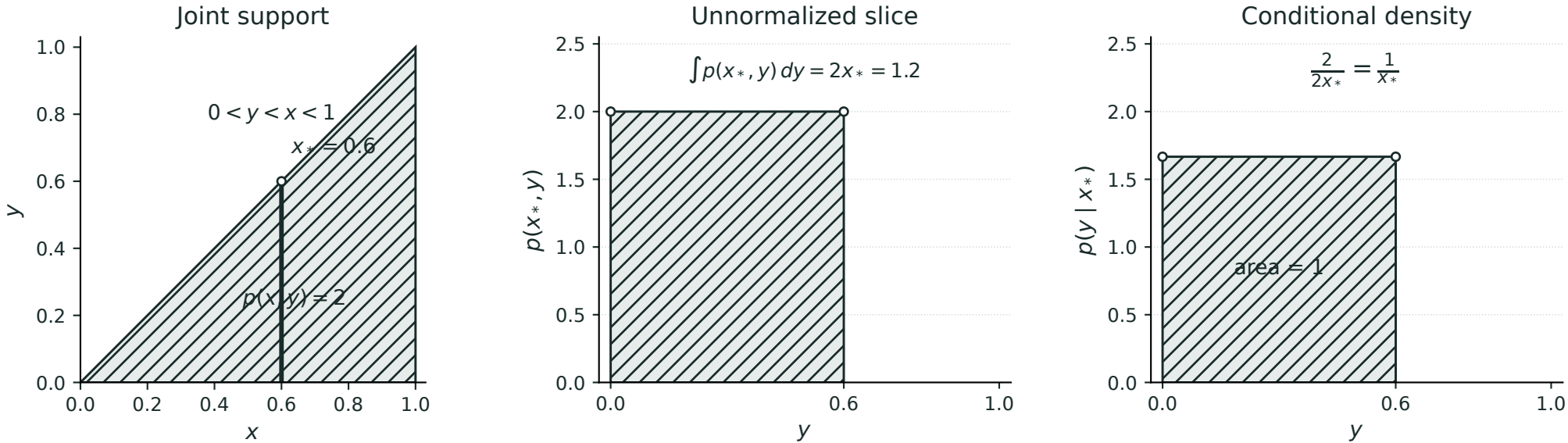
A joint law, its slice, and normalization



For $0 < x < 1$: $p_X(x) = 2x$ and $p(y \mid x) = x^{-1}\mathbf{1}_{(0,x)}(y)$. Boundary values do not determine a conditional version.